
FEASIBLE OPERATING REGION

Multiple aggregators and fairness criteria in the rolling-horizon OPF

Orchestration of the directional sweep, reduction of the computation time
and the resulting mathematical formulation

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Reading convention. Everything that appears **in red** is **new**: it did not exist in the inherited formulation. The rest is reproduced unchanged and is included so that the model is explicit from beginning to end. Part 1 does not alter any equation — it is exclusively orchestration and performance, and its output is bit-identical to that of the previous model.

1 Rolling-horizon orchestration and computation time

The rolling horizon runs through 48 half-hour periods. At each “now” period the day is rotated so that *now* is slot 1 and the **service block** t^* is slot 2; a committed dispatch is solved, it is executed, and 12 directions are swept to trace the region offered at t^* . The formulation did not change: what changed is *what is solved and how many times*.

Both changes were found by profiling the existing sweep, and both are safe only because of how that sweep was already built: the 12 directions genuinely share one feasible region, and the un-rotated inputs are already frozen once per run. What changed is the placement of work that was reasonable when the sweep was written and no longer pays off in the configuration now used in production, a continuous QCP solved by barrier. Reviewed and agreed before pushing.

1.1 The loop, before and after

Before — 25 solves and 13 loads per period

```

for each period  $t$ :
  load DB + Excel
  solve MinImports (1)
  execute slot 1, advance SOC
  for each direction  $\theta_k$ ,  $k = 1..12$ :
    load DB + Excel
    solve MinImportsMaxReverseP
    solve FOR_Vertex( $\theta_k$ )

```

Now — 13 solves and 1 load per period

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load DB + Excel (once per run)
for each period  $t$ :
  light reset
  solve MinImports (1)
  execute slot 1, advance SOC
  for each direction  $\theta_k$ ,  $k = 1..12$ :
    light reset
    solve FOR_Vertex( $\theta_k$ )

```

From one period to the next **one single thing** is carried over: the realised state of charge e_b^0 , which advances with the executed dispatch of slot 1 and comes out of the *baseline*, not out of the sweep. The sweep is handed two: that e_b^0 and the dispatch of slot 1 *fixed*, which guarantees that the offered region is reachable from what the batteries actually do.

1.2 Where the time went

The removed solve, $\min \text{OF}^{\text{imp+rev}}$, ran once per direction. Its objective **does not contain the angle**, and within a period the 12 directions share exactly the same feasible region: same reset, same data rotation, same e_b^0 and the same fixed slot 1. Therefore

The 12 solves of a period were **the same optimization problem** — same matrix, same objective, same solution — solved 12 times. Its purpose was to seed the vertex solve: the answer was kept only as the starting point (*warm start*), and nothing else downstream read it.

1.3 The interventions

- (E) **Splitting the loader.** The reading of the database and Excel was split into a *static* half, which runs once per run, and a per-solve *reset* that restores only the variables that the emptying destroys: the fixing of $q_{b,t}$, the flat voltage seed and the linearization point. Data loads per period: $13 \rightarrow 1$.
- (A) **Removal of the per-direction warm-start solve.** It was suppressed. Solves per period: $25 \rightarrow 13$. It remains behind a switch that restores the previous behaviour, and it is kept *active* in the MIQCP validations, where the start channel is real.
- (C) **Fix in the diagnostic passes.** On splitting the loader, the retries of failed directions stopped inheriting the experimental SOC floor; it is re-applied explicitly so that the retry runs under the same condition as the direction that failed.

1.4 Interior point: why the warm start did not help here

What a *warm start* is. Handing the solver a starting point already close to the answer, so that less ground is left for it to cover. It is standard practice and it usually helps.

Why it did not help here. It depends on the method, and the two available ones look for the optimum in opposite ways:

	Simplex	Barrier (interior point)
Where it advances	along the edge , jumping from vertex to vertex	through the interior , following the central path
Ideal starting point	a vertex close to the optimum — a basis	a well-centred point, far from every active constraint
Effect of starting on the boundary	it brings it closer: few jumps are missing	it penalises it: it must first re-centre

The point that the code handed over was the optimum of *another* problem, the minimum-cost one. And the optimum of a convex program with a linear objective is, by construction, an **extreme point**: stuck against the boundary, with several active constraints. It is exactly what an interior-point method does not want — hence the name of the method: there is a barrier term that pushes it *away* from the constraints, and starting on one of them forces that work to be undone before it can advance. In the corridor image: barrier walks down the middle, and it was being handed the point flattened against the wall.

The change. Suppress that solve. *No equation was touched.* The vertex solve now starts from the flat point that the reset leaves — all voltages at their balanced nominal value —, comfortably interior and with no active constraints, which is just what barrier needs.

Measured result: 1.27 s $\rightarrow$ 0.62 s per vertex solve. *Half, from no longer helping it.*

An honest nuance about the mechanism. What is verified is the *measurement*: 2.05 $\times$ per solve, with bit-identical output. The explanation of the badly centred point is the best supported one, but it has a known tension: on the continuous path the solver does not receive a basis — that channel exists only on the integer path — and the first solve, also a barrier one, does not produce one to hand over either. That the difference is measured all the same indicates that something of the previous state has an influence through a route that was not identified. The *what* is proven; the *why* is a supported hypothesis, not a verified fact.

Note that, even accepting the warm-start theory, **11 of the 12 solves were redundant anyway**: it was enough to solve it once per period and reuse the point. Both programs declare the same sets of constraints and variables, so that they generate a row-for-row identical matrix and differ only in the objective column — which is why removing one does not alter the other.

Numerical options kept: feasibility tolerance 10^{-3} , barrier convergence tolerance 10^{-2} , aggressive scaling, and the compact generator of quadratic programs — needed because the inherited generator expanded the quadratics and exhausted the memory towards period 37 of 48.

1.5 Measured results

	Before	Now	Factor
<i>Micro network — TR9_two, 12 batteries, 48 periods $\times$ 12 directions</i>			
Total time	41 m 22 s	10 m 12 s	4.06×
Vertex solves	731.0 s	358.7 s	2.05×
Per solve	1.27 s	0.62 s	
<i>Production — TR9, 90 prosumers</i>			
Total time	~28 h	3 h 40 m	~ 7.6×
Vertex solves	30 213.4 s	8 440.7 s	3.58×
Per solve	52.45 s	14.65 s	

Equivalence verified. $\max |\Delta \text{proj}| = 0.000\text{e}+00$ in all four runs; 576/576 optimal vertices; byte-identical state-of-charge file; the six metrics of the deliverable reproduced exactly. The acceptance criterion is the **projection** $\theta^\top(P, Q)$, not the equality of (P, Q) : a flat face perpendicular to the sweep direction turns the argument of the maximum into a segment, and the returned point may legitimately slide along it.

The real bottleneck was not the data reading but the **generation of the mathematical program**: on TR9 each generation of a QCP with 90 batteries and 341 buses costs of the order of 30 s, and (A) removes 12 generations per period on top of 12 solves.

2 Formulation: multiple aggregators and fairness at two levels

Unbalanced three-phase current-injection model, with linearization of the load and generation injections around a point $(v_0^{\text{re}}, v_0^{\text{im}})$ and exact quadratic constraints for maximum voltage and inverter capacity. Everything in red is new.

2.1 Sets and indices

$\mathcal{N}$	buses n	$\mathcal{B}$	batteries / prosumers b
$\mathcal{L}$	lines ℓ	$\mathcal{F} = \{1, 2, 3\}$	phases f, h
$\mathcal{K}$	transformers k	$\mathcal{H} = \{1, \dots, 48\}$	periods t
$\mathcal{D}$	loads with PV d	$\mathcal{A}$	aggregators α

$t^\star$ denotes the service block; in the rolling horizon $t^\star = 2$ after the rotation. $a(b) \in \mathcal{A}$ is the aggregator of prosumer b , an input datum.

2.2 Objective function of the directional sweep

For each direction $\theta_k = 30^\circ(k - 1)$, $k = 1, \dots, 12$:

$$\min -(\cos \theta_k P_{t^\star} + \sin \theta_k Q_{t^\star}) \quad (1)$$

with the power at the connection point, positive from grid towards feeder,

$$P_t = \sum_g v_{\text{pcc},f(g)}^{\text{re}} i_{g,t}^{\text{re}} + v_{\text{pcc},f(g)}^{\text{im}} i_{g,t}^{\text{im}} \quad (2)$$

$$Q_t = \sum_g v_{\text{pcc},f(g)}^{\text{im}} i_{g,t}^{\text{re}} - v_{\text{pcc},f(g)}^{\text{re}} i_{g,t}^{\text{im}} \quad (3)$$

Minimising $-\theta_k^\top(P, Q)$ evaluates the *support function* of the feasible set in the direction θ_k ; by rotating θ the boundary of the region is traced.

2.3 Network

The voltage is anchored at the reference bus r , fixed and balanced, and it is what sets the origin of the whole system:

$$v_{r,f,t}^{\text{re}} = \begin{cases} 1 & f = 1 \\ -\frac{1}{2} & f \in \{2, 3\} \end{cases} \quad v_{r,f,t}^{\text{im}} = \begin{cases} 0 & f = 1 \\ -\frac{\sqrt{3}}{2} & f = 2 \\ +\frac{\sqrt{3}}{2} & f = 3 \end{cases} \quad (4)$$

Current balance at each bus and phase:

$$\sum_{g \in n, f} i_{g,t}^{\text{re}} + \sum_{\ell: \text{to}(\ell)=n} i_{\ell,f,t}^{\text{re}} + \sum_{k: \text{to}(k)=n} i_{k,f,n,t}^{\text{re}} - \sum_{\ell: \text{from}(\ell)=n} i_{\ell,f,t}^{\text{re}} - \sum_{k: \text{from}(k)=n} i_{k,f,n,t}^{\text{re}} = \sum_{d \in n, f} (i_{d,t}^{\text{re,L}} - i_{d,t}^{\text{re,G}}) + \sum_{b \in n, f} i_{b,t}^{\text{re}} \quad (5)$$

and its analogue in the imaginary part. Voltage drop on lines, with mutual coupling:

$$v_{\text{from}(\ell),f,t}^{\text{re}} - v_{\text{to}(\ell),f,t}^{\text{re}} = \sum_{h \neq f} (R_\ell^m i_{\ell,h,t}^{\text{re}} - X_\ell^m i_{\ell,h,t}^{\text{im}}) + R_\ell^s i_{\ell,f,t}^{\text{re}} - X_\ell^s i_{\ell,f,t}^{\text{im}} \quad (6)$$

$$v_{\text{from}(\ell),f,t}^{\text{im}} - v_{\text{to}(\ell),f,t}^{\text{im}} = \sum_{h \neq f} (X_\ell^m i_{\ell,h,t}^{\text{re}} + R_\ell^m i_{\ell,h,t}^{\text{im}}) + X_\ell^s i_{\ell,f,t}^{\text{re}} + R_\ell^s i_{\ell,f,t}^{\text{im}} \quad (7)$$

Dy transformers with tap ratio $\tau_{k,t}$, coupling phase f with the next two:

$$v_{\text{from}(k),f,t}^{\text{re}} = -\frac{\nu_k \tau_{k,t}}{3} \left(2v_{\text{to}(k),f\oplus 1,t}^{\text{re}} + v_{\text{to}(k),f\oplus 2,t}^{\text{re}} + R_k (2i_{k,f\oplus 1,t}^{\text{re}} + i_{k,f\oplus 2,t}^{\text{re}}) - X_k (2i_{k,f\oplus 1,t}^{\text{im}} + i_{k,f\oplus 2,t}^{\text{im}}) \right) \quad (8)$$

The v^{im} equation is the same with R_k acting on the imaginary currents and $+X_k$ on the real ones. The primary–secondary current relation is $i_{k,f,\text{from},t}^{\text{re}} \nu_k \tau_{k,t} = i_{k,f,\text{to},t}^{\text{re}} - i_{k,f\oplus 1,\text{to},t}^{\text{re}}$, and analogous in i^{im} . In transformers without on-load tap changing the tap is fixed, $\tau_{k,t} = \tau_k^{\text{fix}}$, so that the product $\tau_{k,t} v$ introduces no nonlinearity: the model remains affine in the injections.

Voltage limits — upper exact, lower on the first-order approximation $\widehat{V}^2$ — and thermal limits:

$$(v_{n,f,t}^{\text{re}})^2 + (v_{n,f,t}^{\text{im}})^2 \leq \overline{V}_n^2, \quad \widehat{V}_{n,f,t}^2 \geq \underline{V}_n^2 \quad (9)$$

$$(i_{\ell,f,t}^{\text{re}})^2 + (i_{\ell,f,t}^{\text{im}})^2 \leq \overline{I}_\ell^2, \quad (i_{k,f,\text{from},t}^{\text{re}})^2 + (i_{k,f,\text{from},t}^{\text{im}})^2 \leq \overline{I}_k^2 \quad (10)$$

where $\widehat{V}_{n,f,t}^2 = (v_0^{\text{re}})^2 + (v_0^{\text{im}})^2 + 2v_0^{\text{re}}(v^{\text{re}} - v_0^{\text{re}}) + 2v_0^{\text{im}}(v^{\text{im}} - v_0^{\text{im}})$. The load and PV injections are linearized in the same way, around $(v_0^{\text{re}}, v_0^{\text{im}})$:

$$i_{d,t}^{\text{re,L}} = \frac{P_{d,t}^0 v_0^{\text{re}} + Q_{d,t}^0 v_0^{\text{im}}}{(v_0^{\text{re}})^2 + (v_0^{\text{im}})^2} + \frac{\partial i^{\text{re,L}}}{\partial v^{\text{re}}} (v^{\text{re}} - v_0^{\text{re}}) + \frac{\partial i^{\text{re,L}}}{\partial v^{\text{im}}} (v^{\text{im}} - v_0^{\text{im}}) \quad (11)$$

2.4 Batteries

With $p_{b,t}$ the net active power of prosumer b ($p_{b,t} > 0$ charging, $p_{b,t} < 0$ discharging) and $q_{b,t}$ the reactive power of the inverter:

$$p_{b,t} = p_{b,t}^{\text{ch}} - p_{b,t}^{\text{dis}}, \quad p_{b,t}^{\text{ch}}, p_{b,t}^{\text{dis}} \in [0, \overline{P}_b] \quad (12)$$

$$p_{b,t} = v_{0,b,t}^{\text{re}} i_{b,t}^{\text{re}} + v_{0,b,t}^{\text{im}} i_{b,t}^{\text{im}}, \quad q_{b,t} = -v_{0,b,t}^{\text{re}} i_{b,t}^{\text{im}} + v_{0,b,t}^{\text{im}} i_{b,t}^{\text{re}} \quad (13)$$

$$e_{b,t} = e_b^0 + \frac{\Delta t}{E_b} \sum_{\tau \leq t} \left(\eta_b^c p_{b,\tau}^{\text{ch}} - \frac{p_{b,\tau}^{\text{dis}}}{\eta_b^d} \right), \quad \underline{\sigma}_b \leq e_{b,t} \leq \overline{\sigma}_b \quad (14)$$

where $e_{b,t}$ is the state of charge as a fraction and $\underline{\sigma}_b, \overline{\sigma}_b$ its bounds — distinct from the capacity E_b , which appears in the denominator and is in units of energy. Inverter capacity at the service block (exact quadratic cone) and cyclic return of the state of charge over the window:

$$p_{b,t^\star}^2 + q_{b,t^\star}^2 \leq \overline{P}_b^2, \quad \left| \sum_{t \in \mathcal{H}} \left(\eta_b^c p_{b,t}^{\text{ch}} - \frac{p_{b,t}^{\text{dis}}}{\eta_b^d} \right) \right| \leq \beta E_b / \Delta t \quad (15)$$

Reactive power is released only at $t^\star$: $q_{b,t} = 0$ for $t \neq t^\star$. In the sweep, the dispatch of slot 1 is fixed to the committed value: $p_{b,1}^{\text{ch}} = \hat{p}_b^c, p_{b,1}^{\text{dis}} = \hat{p}_b^d$. The model also keeps the inherited constraint $p_{b,t^\star} \leq \kappa \overline{P}_b$, where $\kappa = 0$ reproduces the assumption of the paper — batteries only discharging at the service block — and $\kappa = 1$ is the production configuration, bidirectional; with $\kappa = 1$ the constraint is dominated by $p_{b,t^\star}^{\text{ch}} \leq \overline{P}_b$ and is not binding.

2.5 Fairness: proportional criterion C1

New. Let s_b be the nominal capacity of prosumer b — $s_b = \overline{P}_b$ is adopted, which in this model is at once the power of the battery and the radius of the circle of inverter capacity. Let $\zeta \in \mathbb{R}$ be a **single free scalar per solve**, interpretable as the *utilisation factor* of the fleet. With $m \in \{0, 1, 2\}$ the fairness level and $\chi \in \{0, 1\}$ the activation gate:

$$p_{b,t^*} = \zeta s_b \quad \forall b \in \mathcal{B}, \quad \text{if } m = 1 \text{ and } \chi = 1 \quad (16)$$

$$\sum_{b: a(b)=\alpha} p_{b,t^*} = \zeta \sum_{b: a(b)=\alpha} s_b \quad \forall \alpha \in \mathcal{A}, \quad \text{if } m = 2 \text{ and } \chi = 1 \quad (17)$$

A single degree of freedom. The criterion is written with *one* scalar ζ , which removes *one* degree of freedom. Writing it as $n - 1$ pairwise equalities would remove $n - 1$ and would overestimate the cost of fairness by a factor close to two.

Scope of each level. Level 2 constrains only the sums per aggregator; the internal split of each aggregator remains free. The levels are alternative, not cumulative.

Consequence of the sign. Since $p_{b,t^*} = p^{\text{ch}} - p^{\text{dis}}$, a single ζ forces the whole fleet to the same sign: *it forbids one battery to charge while another discharges*. It is the strictest reading of the criterion, and that is why C1 gives the **upper bound** on the cost of fairness.

Feasibility. $\zeta = 0$ is always admissible, so that C1 *contracts* the region and never makes it infeasible. With $s_b = \bar{P}_b$ the implied bound is $|\zeta| \leq 1$.

2.6 The gate χ

χ takes the value 1 in the 12 directional solves and 0 in the solve of the committed dispatch. The reason is not one of implementation:

t^* is slot 2 *also* in the baseline, so that without the gate (16)–(17) would constrain the dispatch executed in slot 1, the trajectory of the state of charge would diverge from the case without fairness, and the comparison would measure two mixed effects: the contraction of the region, and a different battery state. Moreover the objective of the baseline contains no service term at t^* — by construction $w_2(t^*) = 0$ — so imposing a market rule there would constrain an agent that obtains no value at all from participating in it.

2.7 Cases and nesting

m	Case	Constraint active at t^*
0	No fairness (reference)	— no row generated
1	Proportional between prosumers	(16)
2	Proportional between aggregators	(17)

With $m = 0$ both domains are empty: no row is generated and ζ remains as a free column with no coefficients, which presolve removes. That is the reason why the reference case reproduces the previous model exactly.

Nesting. Level 1 implies level 2: by summing (16) over the prosumers of an aggregator, (17) is obtained with the same ζ . Therefore

$$\mathcal{X}(m=1) \subseteq \mathcal{X}(m=2) \subseteq \mathcal{X}(m=0) \implies \text{proj}_{\theta}^{(1)} \leq \text{proj}_{\theta}^{(2)} \leq \text{proj}_{\theta}^{(0)} \quad \forall \theta, \forall t \quad (18)$$

which is the acceptance test: a constraint cannot enlarge the feasible set, and any violation is a defect and not a result.

2.8 Results on the micro network

Full verification on the 576 vertices: the reference case reproduces the previous model with exact $\Delta = 0$ and byte-identical state of charge; the nesting holds without exception; and the constraint is satisfied with a maximum residual of 10^{-4} , an order stricter than the solver's own feasibility tolerance.

	$m = 1$ prosumers	$m = 2$ aggregators
Vertices with contraction	297 / 576 (51.6 %)	272 / 576 (47.2 %)
Maximum projection loss	38.3 kW	16.1 kW
Area loss — median	22.2 %	7.5 %
Area loss — maximum	87.3 %	23.1 %
Maximum $ \zeta $	1.000	1.000

Two observations. (i) The loss of *area* is much larger than that of projection per direction, whose median is practically nil: where the boundary has a flat face perpendicular to the sweep direction, the optimum slides to a fair point at no cost in that direction. Both metrics must be reported. (ii) Scaling a two-dimensional region by γ scales its area by γ^2 : an area loss of 87 % is equivalent to a linear shrinkage to ~ 36 %, not to 13 %.

The fleet is homogeneous — $\bar{P}_b = 3.6$ kW in the 1,216 batteries — so that on this data the proportional criterion coincides numerically with the egalitarian one. That does not make it innocuous: the network is radial, and two prosumers with the same battery at different electrical positions cannot inject the same amount without producing different voltage rises. The heterogeneity that makes C1 bind is **locational**, not equipment.